

REPROCESSING OF SUOMI NPP VIIRS SENSOR DATA RECORDS AND IMPACTS ON ENVIRONMENTAL APPLICATIONS

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ABSTRACT

This study describes the improved calibration applied for VIIRS SDR reprocessing. At reflective solar bands (RSBs), the earth-view F-factor is adjusted with the lunar observations and changes about 2% in band M2-M4 in past five years. An empirical method is applied to remove the radiance spikes in VIIRS thermal emissive bands (TEBs) during the blackbody Warm-Up-Cool-Down (WUCD). VIIRS DNB calibration is improved by applying continuous spectral response changes, new bias tables, updated gain ratio tables, stray light tables, and terrain corrections.

Index Terms— VIIRS, Reprocessing, Reflective Solar Bands (RSB), Thermal Emissive Bands (TEB), Day Night Band (DNB)

1. INTRODUCTION

Suomi NPP (S-NPP) satellite was launched October 2⁸, 2011 and carried VIIRS as one of major payloads [1] [2]. Inheriting from the historical sensors such as NOAA AVHRR, NASA MODIS and Defense Meteorological Satellite Program (DMSP) Operational Linescan System (OLS), the VIIRS covers a spectral range from 0.412 μ m to 11.8 μ m with 14 RSBs, 7 TEBs, and a DNB [3]. To cover the entire Earth during a 24 hour period from a nominal altitude of 829 km, VIIRS has a wide swath field scanning across a 112.56 degree at a spatial resolution of 375 meters for the Imaging (I) bands and 750 meters for the Moderate resolution (M) bands. From VIIRS Sensor Data Records (SDRs), 20 Environmental Data Records (EDRs) are derived

with different sequences and priorities defined by Joint Polar Satellite System (JPSS) Level 1 Requirements Documents (LIRD) [4].

The S-NPP VIIRS was activated on November 8, 2011 and the Earth View (EV) nadir door was opened on November 21, 2011 for the RSB and DNB observations. For TEB EV observations and calibration, the cryo-cooler was opened on January 18, 2012 and the VIIRS became thermally stable and functional after couple of days later. While VIIRS RSB, TEB and DNB calibrations showed better performances when compared to the legacy sensors

such as AVHRR or MODIS [1], there were many calibration and operational changes during the last 5 years since the launch. This study documents the major improvements produced by the NOAA S-NPP VIIRS reprocessing, as compared to the current products generated from the Interface Data Processing Segment (IDPS).

For understanding the source of calibration differences in the reprocessing aspects, we briefly introduce the VIIRS observational sequence from Earth View (EV) to calibration targets including Blackbody (BB), Solar Diffuser (SD), Solar Diffuser Stability Monitor (SDSM) and Space View (SV). In conjunction with calibration schemes, the calibration issues affecting the environmental data records their mitigation methodologies are provided. The overall merits of VIIRS reprocessed are summarized.

2. VIIRS RSB, TEB, AND DNB CALIBRATION

VIIRS radiometric calibration requires an accuracy of 2% in reflectance at the specific radiance value which is dependent RSB band. For meeting this requirement, the VIIRS observes on-board calibrators of SD and SDSM as primary calibration sources. When the VIIRS goes into the termination point from night to day orbit near the South Pole, the SD provides valid SD observations with proper Sun illumination level on the SD panel as shown in Figure 1.

The on-orbit Bidirectional Reflectance Distribution Function (BRDF) of the SD was measured by a series of spacecraft level yaw-maneuvers on February 15 to 16, 2012. These measurements replaced the prelaunch BRDF for the on-orbit RSB calibration [6]. The SD degrades over time at different wavelengths with different rates and the degradation is measured by SDSM as a time-dependent ratio of SD bias-removed Digital Counts (DC) to the Sun DC values [5]. Once the SD degradation, i.e. H-factor, is defined, the VIIRS radiometric calibration coefficient, F-factor, is properly calculated.

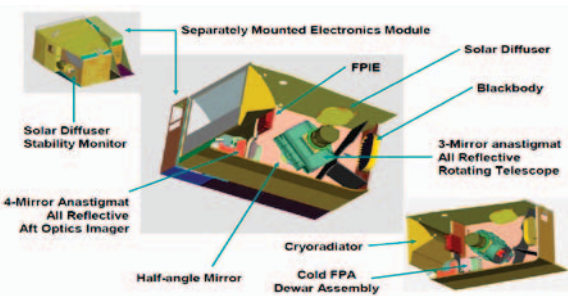


Figure 1. Simplified cross-sectional schematics of the Suomi-NPP VIIRS.

The VIIRS TEB calibration is based on the BB and SV observations, which provide the difference between EV and SV along with the instrument self-emission. The BB emissivity is accurately measured from the prelaunch testing. Considering all the possible components of the in-band spectral response function with the Planck function from the accurately measured BB temperature, blackbody reflected radiance and background radiance from RTA and HAM with RVS correction, the TEB calibration coefficients are calculated [5]. In a normal operation condition, the BB is set to be at 292.5K which provides one radiance level as a calibration point. The BB Warm-Up-Cool-Down (WUCD) cycles are performed every three months to measure the linearity characteristic changes in the calibration coefficients.

DNB calculation is very similar to the RSB calibration except the fact that DNB band has a broad spectral range from 0.45 to 1 μm roughly. The DNB has four different detector arrays of Low Gain Stage (LGS), Mid Gain Stage (MGS) and High Gain Stage (HGS) A and B. The DNB radiometric calibration is primarily calculated by the LGS SD observations. The complexity of the LGS gain calculation is caused by the time-dependent spectral response changes over time, which is caused by the RTA mirror contamination by Tungsten [7]. Another challenging task for DNB calibration is to determine the time-dependent EV dark offset and its drift. It was previously estimated from dark Pacific Ocean observations during a new moon. The dark ocean works well for LGS and MGS but not for HGS because of its high sensitivity of the detectors and air glow in the atmosphere. The dark offset (DN0) is newly derived from the deep space pitch maneuver data which are not affected by the air glow and onboard BB observation data.

3. RESULTS OF VIIRS REPROCESSING

In terms of the three distinctive groups of the RSBs, TEBs and DNB, reprocessing improvements are discussed in this section including a chart comparing with the current operational products. In addition to the estimation of the changes, science community raised issues are briefly

described and the mitigation methodologies of the issues are also discussed..

3.1. RSB issues and improvements

One of the major issues is that the RSB F-factor was suddenly updated two times on May 23 and July 11 in 2014. These sudden updates are caused directly by the SD degradation update because of the anomalous curvature changes in the H-factor around these times [8] [9]. The current IDPS operational H and F-factors are monitored by NOAA Integrated Calibration and Validation System (ICVS) at (<http://www.star.nesdis.noaa.gov/icvs/>). Figure 2 shows current IDPS operational F-factors in red or pink colors and newly proposed Ocean Color (OC) group's version of Look-Up-Table (LUT).

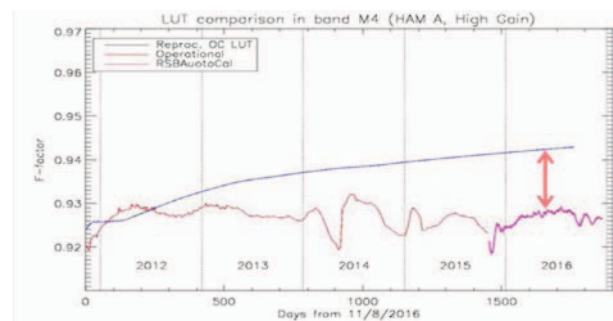


Figure 2. Operational vs. reprocessing F-factor in band M4

In Figure 2, F-PREDICTED LUTs (shown as red-solid line) were replaced by the RSBAutoCal LUTs (pink-solid line) on November 30, 2015. It is clearly seen that the reprocessing OC F-factor (in blue solid line) LUTs are smoother than operational LUT trends with annual oscillations and sudden H/F-factor changes in 2014.

The moon has been used as an independent source of RSB calibration [10]. In NOAA SDR team's recent report, the SD F-factors at RSB bands are mostly consistent with the lunar ones [11][12]. Although the estimated differences in the lunar and SD F-factors are still within the 2% uncertainty requirements [1], specific users such as the ocean color science team request that VIIRS RSB calibration be further improved. Figure 3 shows the NOAA OC team's water leaving radiance products using the SDR product from IDPS (red color) and from the ocean color team (green color [13]). It is clearly seen that the water leaving radiances show a stable trend.

To mitigate these issues in the short wavelength ranges from band M1 to M4, the OC team provided their version of the VIIRS reprocessing F-factor LUTs with the long-term lunar corrections.

Figure 2 shows the band M4 long-term differences between the OC delivered F-factor LUTs and the operational versions. The band M4 differences are rapidly changing in

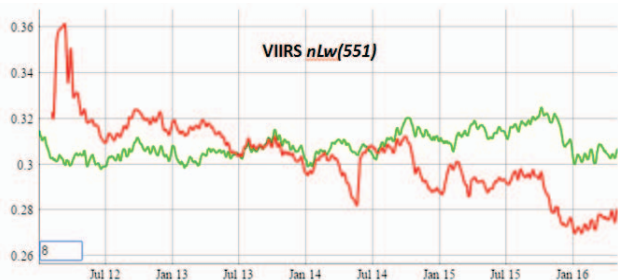


Figure 3. NOAA OC team's water leaving radiance with MOBY in situ data [13].

2012 and gradually increased over time. These long-term differences appearing mostly in bands M1~M4 result from the OC team's lunar correction and the details of the lunar corrections are well described in Sun's recent study [14]. With the lunar correction implemented in the VIIRS SDR reprocessed data, NOAA OC team's water leaving radiance issues are expected to be resolved.

3.2. TEB issues and improvements

VIIRS TEBs have been performing very well since its launch. However, calibration anomaly during WUCD has been observed, and it has small impacts on daytime SST product during such events. The NOAA SST group reported a periodic "global warming" of SST (on the order of 0.3 K) with the S-NPP VIIRS derived SST products during WUCD as shown in Figure 4. To address this issue, an empirical-based correction algorithm (Ltrace) has been developed based on an in-depth analysis of the operational algorithm (Cao 2016, VIIRS SDR teams telecom). It can effectively remove WUCD F-factor anomalies in M15 and M16. The algorithm was validated using collocated CrIS observations and the SST algorithm during the September 19-21 WUCD event. Figure 5 compares global daily-averaged daytime SST anomalies using the operational and reprocessed M15 and M16 SDRs. WUCD anomaly in SST was significantly reduced after the correction. The correction algorithm was implemented in the reprocessing software code.

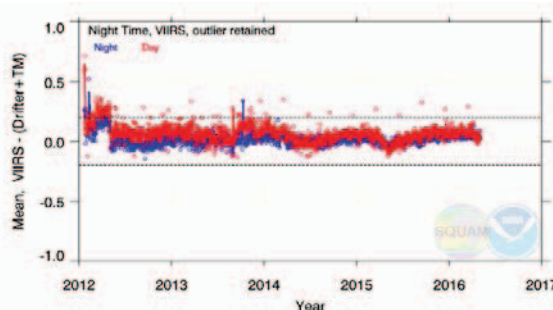


Figure 4. NOAA SST team's global SST bias between VIIRS SST and in-situ measurement [14].

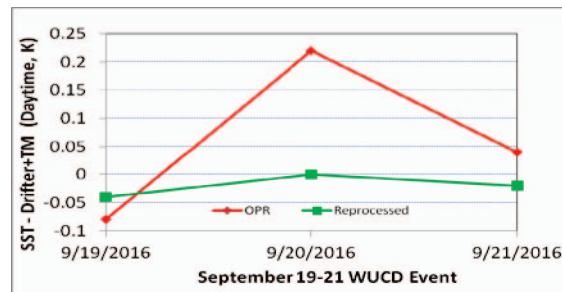


Figure 5. Global daily-averaged SST anomalies using the operational (red color) and reprocessed (green color) M15 and M16 SDRs .

3.3. DNB issues and improvements

The DNB Relative Spectral Response (RSR) is changing continuously over time due to the RTA Tungsten degradation. Current NOAA IDPS system uses only one time update of DNB RSR LUT as of April 5, 2013. The newly developed DNB LGS gain includes the time dependent RSRs thus significantly improved the SDR quality. In addition, the IDPS VIIRS products before March 20, 2012 have a poor radiometric calibration due to the absence of the on-orbit based dark offset and gain ratio. The newly developed on-orbit based offset and gain ratio tables will ensure the better quality of VIIRS SDR products as shown in Figure 6. Such as, the IDPS DNB SDR at night consists of large number of negative radiance values especially during new moon nights. Reprocessed SDR using pitch maneuver based offset helped to reduce the negative DNB radiance values significantly. With the current operational VIIRS products, the DNB stray light correction went operational in August, 2014. An example from correcting the DNB straylight is shown in Fig. 7. Straylight contamination in most of the areas was clearly removed. There are significant qualitative improvements with the straylight correction of the VIIRS DNB granules before August, 2014.

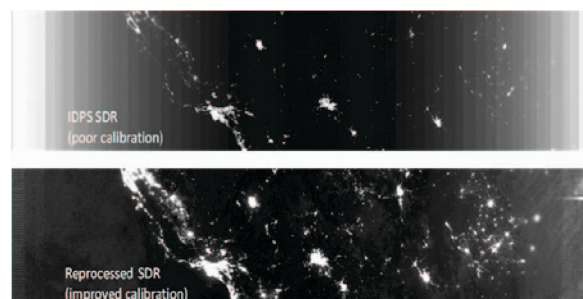


Figure 6. Before and after the reprocessing of a VIIRS DNB granule on March 20, 2012 with the improved gain ratio LUT.

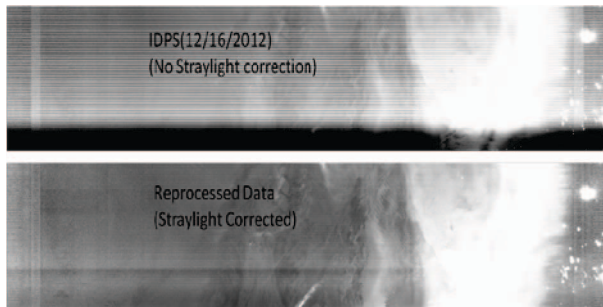


Figure 7. Before and after of the DNB straylight application of a DNB granule acquired on Dec. 16th, 2012

4. SUMMARY

This paper briefly summarized the major calibration issues that affect the science community and that are mitigated in reprocessing. In RSB bands, the new F-factors removed operational sudden changes caused by H-factor updates in 2014 and smoothed its annual oscillation. For the ocean color generated from IDPS, long-term biases in the short-wavelength bands (M1~M4) are corrected by adopting the lunar F-factor trends. The new RSB F-factors also effectively removed the early lifetime calibration uncertainties.

With the TEB bands, a new L_{trance} method is applied to the VIIRS SDR granules which removed anomalous F-factor changes during the WUCD cycles. As a result of the stable F-factors, the periodic ‘global warming’ SST anomalies can be removed.

In addition to these major RSB/TEB band issues, there are major improvements in DNB such as implementations of RSR changes in the gain calculations, new DN0 (offset tables), stray-light LUT correction, and DNB terrain correction in geo-location products.

As shown in this paper, the S-NPP VIIRS mission-long reprocessing is necessary not only to improve SDR products but also to provide accurate radiometric input data for EDR production for better research and applications.

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